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MASTER THESIS SPECIFICATION

Is the Best, the Best? Analysing Best-Paper Awardees' Career Trajectories

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January 30, 2026

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1 Title

Is the Best, the Best? Analysing Best-Paper Awardees' Career Trajectories

2 Abstract

This project investigates how early-career recognition in the form of best paper awards relates to the subsequent trajectories of junior authors. Using large-scale bibliometric data from OpenAlex¹ combined with curated award lists, the study will reconstruct publication, citation, and collaboration histories for awardees and for matched non-awardees at comparable career stages. The analysis focuses on differences in post-award productivity, citation impact, disruptiveness, and co-authorship structure, drawing on established indicators from the science-of-science literature. By examining patterns across conferences and subfields, the thesis aims to assess whether best paper awards simply reflect pre-existing excellence or whether they are associated with distinct developmental advantages. The project provides an empirical contribution to our understanding of how early recognition shapes scientific careers in computer science.

3 Background & Motivation

Best paper awards are among the most visible forms of recognition at computer science conferences. They are often treated as signals of exceptional quality or future impact, shaping how early-career researchers are perceived by hiring committees, funders, and potential collaborators. At the same time, science-of-science work on major prizes such as the Nobel Prize, the Fields Medal, and MacArthur Fellowships paints a mixed picture: some studies find that awards are followed by shifts in focus and lower citation impact per paper, especially for mid- and late-career scientists (1; 2), while others report that early recognition is associated with more publications, more citations, and longer active careers (3). This suggests that awards may function both as markers of past excellence and as interventions that reshape scientists' trajectories in more complex ways.

Compared to these high-profile prizes, much less is known about the effects of conference best paper awards, particularly for junior authors, even though conferences are central publication venues in many CS subfields and such awards are granted far more frequently than global prizes. It remains unclear whether these awards simply select researchers who were already on a strong trajectory, or whether they help create additional advantages in terms of productivity, collaboration opportunities, and the ability to produce disruptive contributions. This project addresses that gap by combining large-scale bibliometric data with award records to compare junior best-paper recipients with similar non-awardees. Using network-science methods on co-authorship and citation relations over time, the thesis will examine how early recognition at conferences relates to later productivity, collaboration patterns, and disruptive contributions.

¹<https://openalex.org>

4 Research Questions

This thesis is guided by the following research questions:

1. What aspects of the career trajectories of junior authors who receive best paper awards at computer science conferences differ from those of comparable non-awarded peers in terms of publication productivity and citation impact?
2. To what extent do awardees and non-awardees differ in the disruptiveness of their subsequent work, as measured by disruption and related bibliometric indicators?
3. What differences are observed in trajectories concentrated in specific venues, sub-fields, or career stages, or are they consistent across contexts?

(Optional) What aspects regarding the structure and evolution of co-authorship networks differ between awardees and matched non-awardees over time (e.g., in terms of collaboration diversity and brokerage roles), based on an operational definition of collaboration diversity to be developed in the thesis?

5 Related Work

Science-of-science research highlights that the effects of major prizes on scientific careers are complex. Nepomuceno et al. report that Nobel laureates and MacArthur Fellows tend to receive fewer citations per publication after receiving an award (1), while Borjas and Doran find that Fields Medalists reduce their publication rate and pursue new topics relative to comparable mathematicians (2). In contrast, Zhu et al. show that earlier recognition of scientific excellence is associated with greater subsequent productivity, higher citation impact, and longer career persistence for awardees compared to closely matched non-recipients (3). Recent work further indicates that a small set of early-career features (early citations, institutional prestige, and collaborations with prominent researchers) strongly predicts researchers' long-term impact across fields, suggesting that advantages accumulated in the first years of a career can persist for decades (4).

Beyond major prizes², several studies evaluate structured early-career programmes and awards using bibliometrics. Berge et al. use traditional indicators (publications, grants) together with network-based measures (network growth, interdisciplinarity, international reach, policy impact) to compare junior faculty participating in a mentored career development programme with a matched comparison group, finding that programme scholars expand their publication output and collaboration networks more strongly over time (5).

Wu, Wang, and Evans show that small teams tend to produce more disruptive contributions, while larger teams typically focus on developing established areas (6). Their work operationalises disruptiveness through the CD index, which compares how often subsequent papers cite a focal work without its predecessors versus alongside them. Together, these strands of research motivate a closer examination of conference-level best paper

²https://jeffhuang.com/best_paper_awards

awards for junior computer science authors, which are frequent yet understudied and situated at the intersection of early-career recognition, collaboration structure, and scientific impact. This thesis extends prior work by shifting attention from major prizes and specialised programmes to conference best paper awards and by combining trajectory analysis, disruption measures, and co-authorship indicators for awardees and matched non-awardees.

6 Description of the Task

This thesis investigates how best paper awards at computer science conferences relate to the long-term career development of junior researchers. The work involves both data collection and statistical analysis across several key steps:

- Collecting large-scale bibliometric metadata from OpenAlex, including publications, citations, authorship, venues, and co-authorship links.
- Curating best-paper award recipients from publicly available award lists and linking them to OpenAlex author records.
- Building comparable control groups of non-awardees from the same conferences, years, and career stages.
- Tracking author trajectories over time, measuring productivity, citation impact, disruption indicators, and collaboration patterns.
- Comparing post-award outcomes between awardees and matched controls using statistical methods.
- Interpreting results in light of existing science-of-science literature and discussing implications for early-career recognition.

Operationalization of trajectories: In our case, career trajectories are defined through quantifiable bibliometric indicators. An upward trajectory is characterized by increasing publications per year and growing citation counts in the post-award period relative to pre-award baselines. Productivity is measured as the number of publications per year; citation impact is measured as total citations received per publication, normalized by publication year to account for citation aging. Disruptiveness is captured via the CD index (6; 7), which measures whether a paper’s impact comes from disrupting existing research streams or consolidating them. We believe these metrics reflect bibliometric patterns rather than subjective assessments of text quality or novelty, acknowledging that scientific excellence has multiple dimensions beyond what can be captured by citation-based measures alone.

7 Problem Statement & Objectives

Best paper awards at computer science conferences are widely seen as signals of excellence, yet their concrete effect on the future careers of *junior* researchers is not well understood. Existing science-of-science work has mostly focused on major prizes and senior scientists,

leaving open whether more frequent, field-specific awards at conferences primarily *select* already promising trajectories or actively *shape* them.

The objective of this thesis is to use large-scale bibliometric and network data to:

- Identify junior authors who receive best paper awards and construct comparable control groups of non-awarded peers;
- Quantify and compare post-award development in terms of productivity, impact, and collaboration patterns;
- Explore whether awardees' careers exhibit signs of more diverse or disruptive contributions than those of similar non-awardees.

8 Scope & Definitions

This thesis focuses on junior authors in selected conference venues. A *junior author* is defined as a researcher within five years of their first indexed publication. Best-paper awardees are compared with non-awardee peers from the same venues and publication years, matched by early-career stage. Only cases with reliable longitudinal publication and co-authorship data are included in the analysis.

9 Methods

This project uses quantitative, data-driven methods grounded in scientometrics and network science. The methodological approach consists of:

- **Data acquisition:** Retrieving publication, citation, and co-authorship metadata from OpenAlex using its API and bulk datasets.
- **Award data integration:** Importing and cleaning curated best-paper award lists (ACM, Jeff Huang, and conference websites) and linking them to OpenAlex author IDs via name matching and disambiguation heuristics.
- **Cohort construction:** Identifying junior authors (within five years of first publication) and creating matched non-awardee controls using conference, year, and early-career indicators (e.g., propensity score matching or exact matching on venue and career age).
- **Metric computation:** Computing productivity (papers/year), impact (citations), disruptiveness (CD index following (6)), and network-based measures including collaboration diversity (unique co-authors, co-authorship network clustering) and brokerage roles.
- **Comparative analysis:** Comparing post-award trajectories using two-sample t-tests or Mann-Whitney U tests for differences in median productivity and citations; linear regression models controlling for pre-award metrics; and time-series visualizations showing year-by-year trends. Effect sizes (Cohen's d) will be reported alongside p-values.

- **Sensitivity analysis:** Assessing robustness to alternate definitions of "junior" (e.g., ≤ 3 years vs. ≤ 7 years), different matching strategies, and subfield-specific effects through stratified analysis.

Addressing subjectivity: This study does not attempt to define "groundbreaking" or "excellent" research qualitatively. Instead, it uses established bibliometric proxies:

- **High disruption ($CD > 0.5$):** Papers that redirect citation flows away from prior work
- **High impact (citations):** Papers that accumulate citations faster than field norms
- **High productivity:** Authors maintaining publication rates above career-stage medians

These are *indicators* of influence and novelty, not definitive measures of scientific quality. The analysis focuses on whether awardees differ systematically from controls on these dimensions, acknowledging that excellence has multiple facets not captured by bibliometrics alone.

Python, Pandas, NumPy, SciPy (for statistical tests), and NetworkX³ will be used to compute metrics and perform analysis. All code and data processing steps will be documented in reproducible notebooks following open science practices.

10 Data Sources

The core bibliometric data for this project will come from OpenAlex, an open scholarly metadata database that provides structured information on publications, authors, venues, and citation links at large scale. Using OpenAlex, the study will construct author- and paper-level panels restricted to computer science venues of interest, including publication years, co-authorship relations, and citation counts.

To identify best paper awardees, the study will combine OpenAlex data with manually curated lists of awards from computer science conferences. As a starting point, we plan to use existing compilations such as the ACM overview of best paper awards⁴ and Jeff Huang's best paper awards collection⁵, complementing them with information retrieved directly from conference websites when necessary. These sources will be matched to OpenAlex records to build cohorts of junior awardees and comparable non-awardees for subsequent analysis.

11 Relevant Courses

The following Master's-level courses provide the background required for this thesis:

³<https://networkx.org/documentation/stable/>

⁴<https://www.acm.org/conferences/best-paper-awards>

⁵https://jeffhuang.com/best_paper_awards/

- **Statistical Machine Learning (5 hp)** — foundations for evaluating scientific impact and longitudinal modelling.
- **Natural Computation Methods for Machine Learning (10 hp)** — optimization and computational techniques used in scientometric analysis.
- **Linear Algebra for Data Science (5 hp)** — mathematical tools for embeddings, similarity measures, and network analysis.
- **Introduction to Data Science (10 hp)** — core DS workflows including cleaning, modelling, and evaluation.
- **Data Engineering I (5 hp) and Data Engineering II (10 hp)** — experience with large-scale data handling, pipelines, and distributed computation, essential for processing OpenAlex data.
- **Database Design I (5 hp)** — structured data modelling and querying relevant for bibliometric integration.
- **Software Engineering and Project Management (5 hp)** — contributes to reproducibility, documentation, and project structure.

12 Delimitations

To ensure that the project remains feasible within the time frame, several aspects are explicitly excluded:

- **No qualitative surveys or interviews:** The project focuses solely on quantitative bibliometric data.
- **No analysis of non-CS awards or journal best-paper awards:** Only selected computer science conferences are included.
- **No manual annotation of authors or papers:** All matching will rely on metadata-based heuristics.
- **No development of new machine learning models:** The thesis uses established bibliometric and network-science indicators.
- **Network analysis is optional:** Co-author network measures will be included only if time allows after completing core trajectory analysis.
- **No prediction or causal inference:** the study aims for descriptive and comparative analysis, not forecasting or causal modelling.

13 Ethical and Societal Considerations

This project relies exclusively on publicly available bibliometric metadata from OpenAlex and publicly listed best paper awards. No personal, sensitive, or confidential data is collected. Nevertheless, several ethical and societal considerations are relevant.

Data quality, fairness, and representation

Bibliometric datasets may contain errors, incomplete records, or inconsistent author disambiguation. Such issues can disproportionately affect early-career researchers, researchers from smaller institutions, or those with non-Western names. The study will acknowledge these limitations and avoid drawing conclusions that assume perfect data quality or individual-level precision.

Responsible interpretation of impact indicators

Citation counts, disruption scores, and other impact metrics reflect many factors beyond scientific quality, including visibility, collaboration networks, and systemic biases in the publication ecosystem. The project will treat these indicators as descriptive signals rather than normative measures of scientific worth, and results will be framed accordingly.

Societal impact and academic inequality

Awards and recognition systems are not neutral gatekeepers of merit. By examining whether best-paper awards amplify advantage or distribute opportunity equitably, this thesis investigates a critical mechanism shaping academic careers.

Award committees may have limited visibility into researchers outside prominent institutions; award recipients may convert recognition into career gains (collaborations, funding, publications) more easily if they already occupy well-resourced networks; and award patterns may signal which research topics are valued, steering junior researchers toward institutional expectations rather than bold directions.

We will examine post-award outcomes stratified by institutional context and subfield to detect whether award effects are distributed equitably. Our analysis will remain alert to how bibliometric indicators (citations, disruption scores, productivity) reflect network position and visibility, not intrinsic worth. If disparities emerge, they suggest that award systems may require redesign to support equal opportunity.

Transparency and reproducibility

All analysis will be performed with transparent methodology, documented data-processing pipelines, and reproducible code. No proprietary data or confidential information will be used. Any limitations in data coverage or methodological assumptions will be explicitly reported to avoid misinterpretation of the results.

Use of generative AI

Generative AI tools may be used for language refinement and code assistance, in accordance with the university guidelines and with full transparency in the methodology and final report. Scientific reasoning, data analysis, and interpretation will remain the student's own work.

If no additional ethical concerns arise during the course of the project, this will be explicitly stated in the final report.

14 Project Duration

The project will be conducted during the official course period for *Degree Project E in Data Science (1DL510)*, Spring 2026 Period 1.

- **Start date:** 19 January 2026
- **End date:** 7 June 2026
- **Pace of study:** 100%
- **Language of instruction:** English

These dates correspond to the designated teaching period published for the course, and the project plan and milestones are aligned with this timeframe.

15 Work Plan & Milestones

Table 1: Planned stages, timeline, and main milestones

Stage	Weeks	Task	Deliverable
1	4	Setup: repo, OpenAlex access, define scope	Intro outline; demo notebook
1	5–6	Literature review; award list curation	Background & Related Work draft
1	7–8	Data pipeline; build cohorts (awardees & controls)	Methods section; preliminary stats
2	9–10	Compute metrics (productivity, citations, CD index)	Results: descriptive analysis
2	11–12	Multi-venue analysis; stratify by subfield	Results: comparative analysis
2	13–14	Network analysis (optional); regression models	Results complete; Discussion draft
3	15–16	Sensitivity analyses; robustness checks	Discussion final; Conclusion
3	17–18	Finalize figures; reproducibility docs	Complete first draft
3	19–20	Supervisor feedback; polish writing	Revised full draft
3	21–22	Proofreading; presentation prep	Camera-ready thesis
3	23	Submission (June 7); defense prep	Thesis submitted

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